

Hongwei Li

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1 Research Interests

My research centers around **AI agents for Security and SE and agentic Fine-tuning**. My current research focuses on developing customized *AI agents systems* and *reasoning LLMs* for software security and engineering tasks such as patch generation, vulnerability detection, and PoC generation. I also enjoy playing CTF as a core member of Shellphish, a renowned CTF team.

2 Highlighted Projects & Skills

Advanced LLM agents for software engineering and security

- Design and develop pioneer LLM agents with *domain-specific tools, advanced workflows, and memory managements* for SE and security
- Outcomes:
 - **OpenSage** — A self-programming agent generation engine, ranked Top-1 agent on Terminal-Bench, accepted at ICML 2026;
 - **SageAgent-CTF** — A fully autonomous security agent built on OpenSage, solved 7 challenges in DEF CON CTF Qualifiers 2026;
 - **PatchPilot** — A cost-efficient issue resolving agent, ranked Top-5 tools on SWE-bench-Verified;
 - **Artiphishell** — A multi-agent system for vulnerability detection and patching, ranked Top-5 in DARPA AIXCC competition;
- Skills: Google Agent Development Kit, Agent design, MCP protocols, Fuzzing (Generic/Directed /Kernel/Web3), Static Analysis(CodeQL/Joern), Concolic Execution(Angr), Reverse Engineering, Vulnerability Exploitation

LLM post-training and agentic training

- Design *novel training recipes for distillation with agentic data*; data selection strategies, running multiple teacher models in our domain-specific agents, unique data filtering and constitution-based data rationalization and correction, and SFT-based fine-tuning; the obtained small models outperform large teacher models in our target tasks
- Outcomes:
 - **VulnLLM-R** - The first reasoning-based large language model for vulnerability detection, equipped with agent scaffolding and agentic fine-tuning, outperforms SOTA tools;
 - **Co-PatcheR** — A set of small reasoning models for issue resolving, ranked Top-2 open-weight models on SWE-bench-Verified;
- Skills: LLaMA-Factory, SFT, vLLM, RL, PyTorch, HuggingFace, verl, Data/model/pipeline parallelism (e.g., FSDP, DDP)

3 Education

- University of California, Santa Barbara**, Goleta, CA Sep. 2024 – Present
Ph.D. Student, Department of Computer Science, advised by Prof. Wenbo Guo
- University of California, Berkeley**, Berkeley, CA Feb. 2026 – Mar. 2026
Visiting Affiliate Researcher, Department of Computer Science, hosted by Prof. Dawn Song
- Purdue University**, West Lafayette, IN Sep. 2023 - May. 2024
Ph.D. Student, Department of Computer Science, advised by Prof. Wenbo Guo
- Shanghai Jiao Tong University**, Shanghai, China Aug. 2020 - Mar. 2023
M.Eng. in Cybersecurity, advised by Prof. Yue Wu
- Shanghai Jiao Tong University**, Shanghai, China Sep. 2016 - Jun. 2020
B.Eng. in Information Engineering; B.A. in French (Dual Degree)

4 Publications

LLM post-training and agents

1. [ICML'26] **Hongwei Li**[†], Zhun Wang[†], Qinrun Dai, Yuzhou Nie, Jinjun Peng, Ruitong Liu, Jingyang Zhang, Kaijie Zhu, Jingxuan He, Lun Wang, Yangruibo Ding, Yueqi Chen, Wenbo Guo, Dawn Song, "OpenSage: Self-programming Agent Generation Engine", In Proceedings of *The 43rd International Conference on Machine Learning*, Seoul, KR, 2026
[†] These authors contributed fully equally as co-first authors; author order does not reflect contribution
2. Zhun Wang, Nico Schiller, **Hongwei Li**, Srijiith Sesha Narayana, Milad Nasr, Nicholas Carlini, Xiangyu Qi, Eric Wallace, Elie Bursztein, Luca Invernizzi, Kurt Thomas, Yan Shoshitaishvili, Wenbo Guo, Jingxuan He, Thorsten Holz, Dawn Song, "ExploitGym: Can AI Agents Turn Security Vulnerabilities into Real Attacks?", Under Review, 2026
3. Yuzhou Nie, **Hongwei Li**, Ruizhe Jiang, Chengquan Guo, Zhun Wang, Bo Li, Dawn Song, Wenbo Guo, "VulnLLM-R: Specialized Reasoning LLM with Agent Scaffold for Vulnerability Detection", Under Review, 2026
4. Shellphish, "An AI-powered Cyber Reasoning System for Automatic Vulnerability Identification and Patching", Under Review, 2026
5. Tianneng Shi, Jingxuan He, Zhun Wang, Linyu Wu, **Hongwei Li**, Wenbo Guo, Dawn Song, "Progent: Programmable Privilege Control for LLM Agents", Under Review, 2026
6. Yujin Potter, Wenbo Guo, Zhun Wang, Tianneng Shi, **Hongwei Li**, Andy Zhang, Patrick Gage Kelley, Kurt Thomas, Dawn Song, "SoK: Frontier AI's Impact on the Cybersecurity Landscape", Under Review, 2026
7. [ICLR'26] Yuheng Tang, Kaijie Zhu, Bonan Ruan, Chuqi Zhang, Michael Yang, **Hongwei Li**, Suyue Guo, Tianneng Shi, Zekun Li, Christopher Kruegel, Giovanni Vigna, Dawn Song, William Yang

Wang, Lun Wang, Yangruibo Ding, Zhenkai Liang, Wenbo Guo, “DevOps-Gym: Benchmarking AI Agents in Software DevOps Cycle”, In Proceedings of *The 14th International Conference on Learning Representations*, Rio de Janeiro, Brazil, 2026

8. [ICML’25] **Hongwei Li**, Yuheng Tang, Shiqi Wang, Wenbo Guo, “PatchPilot: A Verifiable and Cost-Efficient Agentic Patching Framework”, In Proceedings of *The 42nd International Conference on Machine Learning*, Vancouver, CA, 2025
9. [NeurIPS’25] Yuheng Tang, **Hongwei Li**, Kaijie Zhu, Michael Yang, Yangruibo Ding, Wenbo Guo, “Co-PatcheR: Collaborative Software Patching with Component-specific Small Reasoning Models”, In Proceedings of *The 39th Conference on Neural Information Processing Systems*, San Diego, CA, 2025

RL and DNN for Security

1. Wenxuan Shi, **Hongwei Li**, Jiahao Yu, Xinqian Sun, Wenbo Guo, Xinyu Xing, “BandFuzz: An ML-powered Collaborative Fuzzing Framework.”, arXiv preprint arXiv:2507.10845.
2. [ICSE SBFT’24] Wenxuan Shi, **Hongwei Li**, Jiahao Yu, Wenbo Guo, Xinyu Xing, “BandFuzz: A Practical Framework for Collaborative Fuzzing with Reinforcement Learning”, In Proceedings of *International Workshop on Search-Based and Fuzz Testing*, Lisbon, Portugal, 2024
3. [Computers & Security’22] Jingcheng Yang, **Hongwei Li**, Shuo Shao, Futai Zou, Yue Wu, “FS-IDS: A framework for intrusion detection based on few-shot learning”, *Computers & Security*, 122: 102899, 2022.

5 Selected Competitions

- Terminal-Bench, 2026 (**Top 1 Agent**) (OpenSage)
- AIXCC, DARPA, 2023-2025 (**5th Place; Awarded \$3M**)
- SWE-Bench patching leaderboard, 2025 (**Top 5 tools; Top 2 Open-weight models**)
- Google SBFT Fuzzing Tool Competition, 2024 (**Top 1, Awarded \$23K**)
- DEF CON CTF Qualifiers, 2026 (**45th Place**) (SageAgent-CTF, fully automated, 7 challenges solved)
- DEF CON CTF Final, 2025 (**11th Place**)
- SunshineCTF, Sep 2021 (**11th Place**) (*Solo Team in a Team-based CTF*)